#BLD23652

Quick Work

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Date:

M T W T F S S

24/03/26

T-8 (Decision Tree)

ID3

↙

Entropy & Information Gain

(IG) → Information Gain of each feature

IG → max → best feature

Entropy ↑ randomness ↑

$$\text{Entropy} \rightarrow -\sum_{i=1}^{N} P_i * \log_2(P_i)$$

$$\log_2 P_i = \frac{\log_{10} P_i}{\log_{10} 2} = \frac{\ln P_i}{\ln 2}$$

$$IG = \text{Entropy (parent)} - \sum \frac{N_k}{N} \text{Entropy}(k)$$

i) Entropy (Parent or Y) = 1.

$$\therefore P_{spam} = \frac{5}{10} = 0.5$$

$$P_{Nspam} = \frac{5}{10} = 0.5$$

$$= -(0.5\log_2(0.5) + 0.5\log_2(0.5)).$$

$$= 1.$$

ii) Entropy (Money keywords) = 0.

1 → spam.

~~2 → Non spam~~

0 → Non spam

— spam

Information Gain (M.K)

= 1 − 0 = 1.

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▷ Entropy (Sender known) =

1 → Spam (1)

→ Non Spam (5)

0 – Spam = 4

Non-Spam = 0.

$$= \left[\frac{1}{6}\log_2\left(\frac{1}{6}\right) + \frac{5}{6}\log_2\left(\frac{5}{6}\right)\right] + 0\,\frac{4}{4}\log_2\left(\frac{4}{4}\right)\Big]$$

$= 0.65$

IG =

$$1 - \left(\frac{6}{10} \times 0.65 + \frac{4}{10} \times 0\right)$$

$= 0.61.$

▷ Entropy (Has Attachments)

0 — Spam = 2, Non-Spam = 3 } Entropy > 0.

1 — Spam = 3, Non-Spam = 2 } Entropy > 0.

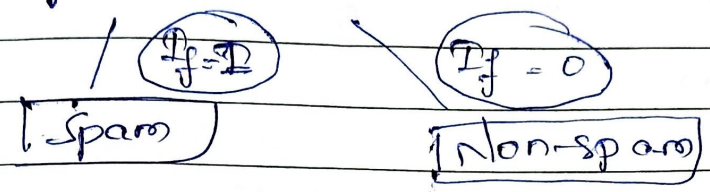
IG < 1

▷ Entropy (Suspecious words) =

Entropy > 0

IG < 1

Money keywords (Root Node).



Depth = 1.

Leaf Nodes = 2